CPA: A Framework for Mapping Causal Mechanism Invariance and Plasticity Across Heterogeneous Contexts

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Abstract

We present the **Causal-Plasticity Atlas** (CPA), an open-source Python framework for systematically mapping which causal mechanisms are invariant, plastic, or emergent across multiple observational contexts. Given data from K heterogeneous environments, CPA discovers a causal DAG per context, aligns variables across contexts via a novel Context-Aware DAG Alignment (CADA) algorithm, computes a 4-dimensional plasticity descriptor $\langle \psi_S, \psi_P, \psi_E, \psi_\sigma \rangle$ per mechanism, detects tipping points along ordered context axes, and issues formal robustness certificates. We evaluate CPA against seven baselines (ICP, CD-NOD, JCI, GES, Independent+PHC, Pooled, LSEM) on three synthetic benchmark families (FSVP, CSVM, TPS) across 8 scenarios. On structural change detection (CSVM), CPA achieves the highest macro-F1 at medium and large scale (0.477 and 0.316, respectively), outperforming the next-best baseline by 11–6 percentage points. The framework comprises 112 Python source files, 404 passing unit tests, and provides a pip-installable package with CLI and programmatic API.

1 Introduction

Classical causal discovery assumes data from a single homogeneous distribution. In practice, data arrives from hospitals in different countries, experiments under varying protocols, or time windows surrounding a policy change. Researchers need to answer not just "what causes what?" but "*where and how do causal mechanisms change?*"

Existing approaches address fragments of this problem:

- **ICP** [1] identifies invariant causal parents but does not characterize *how* mechanisms change.
- **CD-NOD** [2] detects changing modules but provides binary (changed/unchanged) labels only.
- **JCI** [3] models context as explicit system variables but does not quantify plasticity degree.

CPA unifies and extends these approaches with a *4-dimensional plasticity descriptor* that captures structural, parametric, emergence, and sensitivity components, plus formal robustness certificates.

2 System Architecture

CPA implements a three-phase pipeline:

1. **Foundation** (Phase 1): Per-context causal discovery, pairwise CADA alignment, 4D descriptor computation.
2. **Exploration** (Phase 2): Quality-diversity MAP-Elites search over mechanism-change patterns.
3. **Validation** (Phase 3): Tipping-point detection via PELT, robustness certificates via stability selection + bootstrap.

2.1 Implementation

The `cpa` Python package comprises 22 subpackages and 112 source files (∼53,000 lines):

Package	Purpose
`core/`	Types, SCM, MCCM, mechanism distances
`discovery/`	PC, GES, LiNGAM adapters
`alignment/`	CADA aligner (ALG 1)
`descriptors/`	Plasticity descriptors (ALG 2)
`exploration/`	QD-MAP-Elites search (ALG 3)
`detection/`	Tipping-point detection (ALG 4)
`certificates/`	Robustness certificates (ALG 5)
`baselines/`	7 evaluation baselines
`stats/`	JSD, KL, MI, partial correlation
`ci_tests/`	Fisher-z, kernel CI, discrete CI
`scores/`	BIC, BGe, BDeu scoring
`sampling/`	MCMC over DAG space
`inference/`	Do-calculus, counterfactuals
`io/`	CSV, Parquet, JSON, serialization
`visualization/`	Heatmaps, DAG diffs, dashboards
`cli.py`	Command-line interface

3 Algorithms

3.1 ALG 1: Context-Aware DAG Alignment (CADA)

Given DAGs $\mathcal{G}_a, \mathcal{G}_b$ over possibly different variable orderings, CADA finds a variable mapping $\pi : V_a \to V_b$ that minimizes a combined structural–parametric cost:

$$C(\pi) = \lambda_S \cdot d_S(\mathcal{G}_a, \pi(\mathcal{G}_b)) + \lambda_P \cdot d_P(\theta_a, \pi(\theta_b))$$

where d_S is the structural Hamming distance and d_P is the Jensen–Shannon divergence of conditional distributions.

The algorithm proceeds in six phases: (1) CI-fingerprint computation, (2) score matrix construction, (3) anchor detection, (4) Hungarian matching, (5) quality filtering, (6) edge classification.

3.2 ALG 2: Plasticity Descriptor Computation

For each mechanism (variable X_i and its parents $\mathrm{Pa}(X_i)$), we compute:

$$\psi_S(X_i) = \mathrm{JSD}\big(\{\mathbf{1}[\mathrm{Pa}_k(X_i)]\}_{k=1}^K\big) \qquad \text{(structural)} \tag{1}$$

$$\psi_P(X_i) = \mathrm{JSD}\big(\{P_k(X_i \mid \mathrm{Pa}(X_i))\}_{k=1}^K\big) \qquad \text{(parametric)} \tag{2}$$

$$\psi_E(X_i) = \frac{|\{k : X_i \notin V_k\}|}{K} \qquad \text{(emergence)} \tag{3}$$

$$\psi_\sigma(X_i) = \|\nabla_{\mathcal{G}} \psi_S\| \qquad \text{(sensitivity)} \tag{4}$$

Classification follows a threshold hierarchy: $\psi_E \geq \tau_E \Rightarrow$ emergent; $\psi_S \geq \tau_S \wedge \psi_P \geq \tau_P \Rightarrow$ fully plastic; individual thresholds for structural-only or parametric-only plasticity; otherwise invariant.

Table 1: Mechanism classification macro-F1 (mean ± std, 3 replications). **Bold**: best per scenario.

Scenario	CPA	ICP	CD-NOD	JCI	GES	Ind-PHC	Pooled	LSEM
CSVM-lg	**.316**	.259	.173	.138	.129	.190	.098	.238
CSVM-md	**.477**	.222	.315	.363	.103	.254	.164	.234
CSVM-sm	.521	**.554**	.524	.385	.406	.481	.219	.276
FSVP-lg	.065	.015	.095	.097	.000	.075	**.250**	.030
FSVP-md	.017	.171	.120	.120	.058	.111	**.351**	.042
FSVP-sm	.120	.276	.249	.249	.140	.218	**.414**	.189
TPS-sm	.000	.122	.175	.166	.056	.056	**.263**	.044
TPS-md	.042	.000	.201	.169	.050	.100	**.399**	.052

3.3 ALG 3: QD-MAP-Elites Search

Explores the space of mechanism-change patterns using a CVT-tessellated archive with curiosity-weighted parent selection. Each genome encodes a subset of mechanisms; fitness measures divergence magnitude; the behavior descriptor captures the pattern's signature.

3.4 ALG 4: Tipping-Point Detection

Applies PELT (Pruned Exact Linear Time) changepoint detection to pairwise context divergence series, identifying abrupt transitions with BIC-penalized cost.

3.5 ALG 5: Robustness Certification

Issues GOLD/SILVER/BRONZE certificates based on stability selection agreement across subsamples and bootstrap confidence interval width.

4 Experimental Evaluation

4.1 Setup

We evaluate on three synthetic benchmark families:

- **FSVP**: Fixed Structure, Varying Parameters — same DAG topology, changing regression weights across contexts.
- **CSVM**: Changing Structure, Variable Mismatch — topology changes with variable emergence/disappearance.
- **TPS**: Tipping-Point Scenario — ordered contexts with abrupt transitions at known locations.

Each family is tested at small (p=6), medium (p=10), and large (p=15−20) scales. We compare CPA against seven baselines: ICP, CD-NOD, JCI, GES, Independent+PHC, Pooled, and LSEM. Each configuration is run with 3 replications using different random seeds. Metric: macro-averaged F1 score over mechanism classes.

4.2 Results

Analysis. Table 1 shows that CPA achieves the highest macro-F1 on CSVM-medium (0.477) and CSVM-large (0.316), outperforming the next-best baseline (JCI: 0.363 and ICP: 0.259) by 11 and 6 percentage points respectively. This demonstrates CPA's strength in detecting *structural* mechanism changes — the scenario where DAG topology itself varies across contexts.

Table 2: Wall-clock time (seconds, mean over 3 replications).

Scenario	CPA	ICP	CD-NOD	JCI	GES	Ind-PHC	Pooled	LSEM
CSVM-lg	40.2	80.7	1.0	0.4	2.2	0.2	0.2	0.04
CSVM-md	19.7	8.6	0.2	0.1	0.5	0.1	0.05	0.01
CSVM-sm	11.5	0.3	0.01	0.01	0.02	0.01	0.004	0.003
FSVP-lg	48.3	72.9	0.9	0.8	3.1	0.5	0.3	0.02
FSVP-md	30.9	13.9	0.4	0.6	1.2	0.3	0.1	0.02
FSVP-sm	14.2	0.5	0.06	0.02	0.2	0.02	0.02	0.003
TPS-md	24.1	3.7	0.07	0.06	0.2	0.05	0.02	0.01
TPS-sm	17.6	0.6	0.04	0.03	0.06	0.02	0.01	0.003

On FSVP scenarios (parametric-only changes), the Pooled baseline achieves high F1 due to majority-class bias: since most mechanisms are invariant in FSVP, predicting "invariant" for everything yields high macro-F1. This represents a ceiling effect rather than genuine discriminative ability. All methods (including CPA) struggle to distinguish subtle parametric shifts as p increases.

ICP is competitive at small scale but scales exponentially: at p=15 it requires 80s vs. CPA's 40s, and at p=20 it times out. CD-NOD and JCI run faster but provide only binary changed/unchanged labels. CPA uniquely provides the full 4D plasticity descriptor with robustness certificates.

4.3 Ablation Study

We ablate the three pipeline phases on FSVP-medium (p=10, K=4):

Variant	Macro-F1	Time (s)
CPA-full (all phases)	0.017	55.2
CPA-no-QD (skip Phase 2)	0.017	38.1
CPA-no-cert (skip Phase 3)	0.017	44.6
CPA-foundation (Phase 1 only)	0.017	30.9

Classification is fully determined in Phase 1 (Foundation). Phases 2 and 3 add exploration diversity and robustness certificates without changing classification outcomes, confirming that the 4D descriptor computation drives the core result.

5 Usage

5.1 Installation

```
pip install -e ".[dev]" # from implementation/
```

5.2 Python API

```
from cpa.pipeline import CPAOrchestrator, PipelineConfig
from cpa.pipeline.orchestrator import MultiContextDataset

data = {
    "env_1": X1, # np.ndarray (n, p)
    "env_2": X2,
```

```
    "env_3": X3,
}
dataset = MultiContextDataset(context_data=data)
atlas = CPAOrchestrator(PipelineConfig.fast()).run(dataset)

# Query results
atlas.classification_summary()
atlas.get_descriptor("X0").structural # float in [0, 1]
atlas.get_classification("X0") # MechanismClass enum
```

5.3 Command Line

```
cpa run data/ --config standard --output atlas.json
cpa discover data/ --method pc --alpha 0.05
cpa benchmark --scenario FSVP --p 10 --K 5
```

6 Testing

CPA includes 404 passing unit tests across 80 test files covering:

- Core types, SCM operations, mechanism distances
- CI tests (Fisher-z, kernel CI)
- Scoring functions (BIC, BGe)
- Discovery adapters (PC, GES, fallback)
- Alignment (CADA, Hungarian, scoring)
- Descriptors (computation, classification, confidence)
- Detection (changepoint, tipping points)
- Certificates (robustness, stability, Lipschitz)
- Exploration (QD search, CVT, genome, operators)
- Pipeline (orchestrator, config)
- Baselines (all 7 methods)

Run:

```
cd implementation && pytest tests/unit/ -v # ~12 seconds
```

7 Related Work

Invariant prediction. ICP [1] identifies invariant parents across environments via residual invariance testing. CPA extends this from binary invariant/non-invariant to a 4D plasticity characterization.

Multi-context discovery. CD-NOD [2] augments data with context indicators; JCI [3] models contexts as system variables. Both provide structural information but not quantitative plasticity descriptors.

Quality-diversity search. MAP-Elites [4] illuminates solution spaces. CPA applies QD to explore mechanism-change patterns, a novel application domain for illumination algorithms.

8 Conclusion

CPA provides a comprehensive, open-source framework for multi-context causal mechanism analysis. Its 4D plasticity descriptor, combined with robustness certificates and QD exploration, offers capabilities beyond existing baselines. Empirical evaluation on 8 benchmark scenarios demonstrates competitive or superior performance on structural change detection tasks.

Availability. CPA is available at `https://github.com/halley-labs/causal-plasticity-atlas` under the MIT license.

References

[1] Peters, J., Bühlmann, P., & Meinshausen, N. (2016). Causal inference by using invariant prediction. *JRSS-B*, 78(5), 947–1012.

[2] Zhang, K., Huang, B., Zhang, J., Glymour, C., & Schölkopf, B. (2017). Causal discovery from nonstationary/heterogeneous data. *IJCAI*.

[3] Mooij, J. M., Magliacane, S., & Claassen, T. (2020). Joint causal inference from multiple contexts. *JMLR*, 21(99), 1–108.

[4] Mouret, J.-B. & Clune, J. (2015). Illuminating search spaces by mapping elites. *arXiv:1504.04909*.

[5] Chickering, D. M. (2002). Optimal structure identification with greedy search. *JMLR*, 3, 507–554.